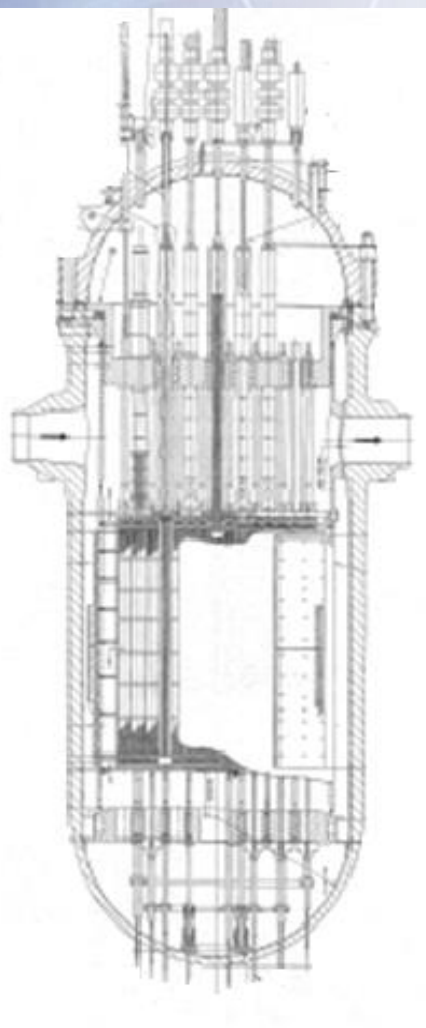
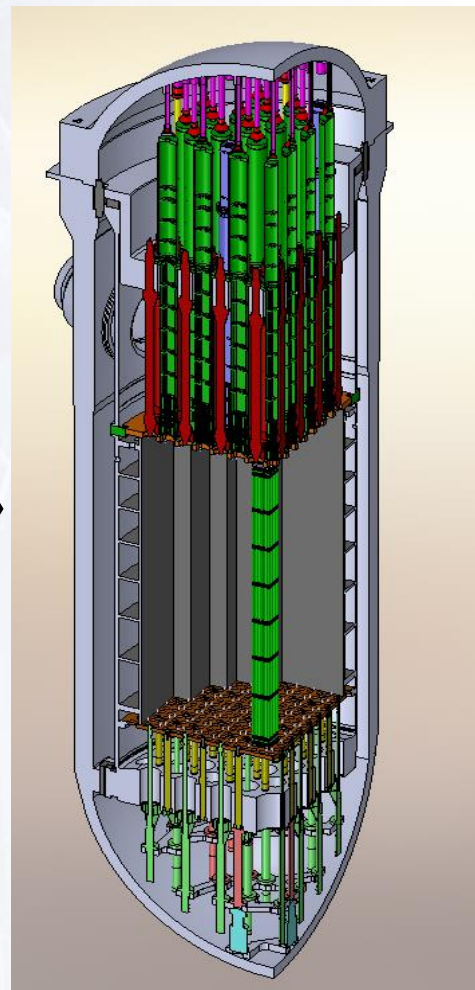


4-Loop Westinghouse PWR Multi-Physics Model Development

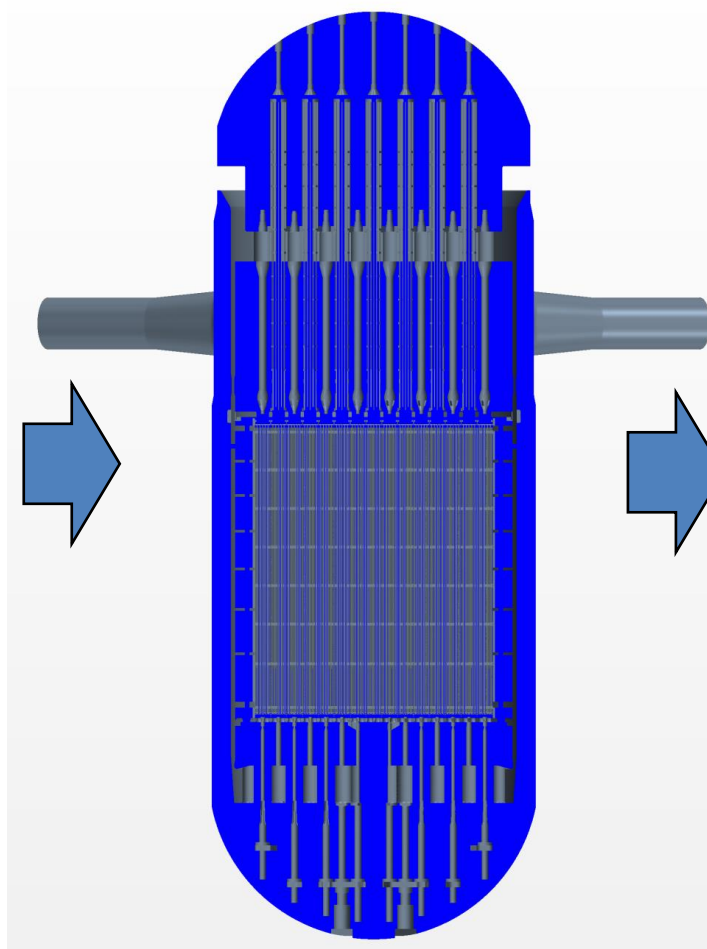
- RPV ID 173", 193/4 Fuel Assemblies, 13,944 fuel rods (fuel pellets, helium gap), 434 spacers, 148,224 mixing vanes; 1.2 billion cells



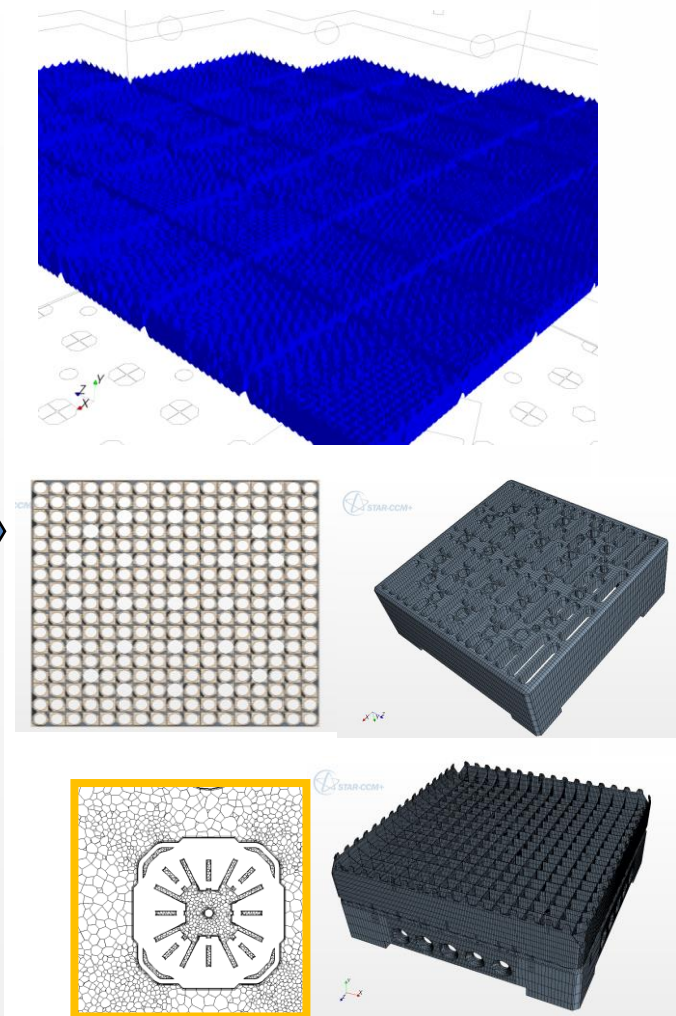
DRAWINGS



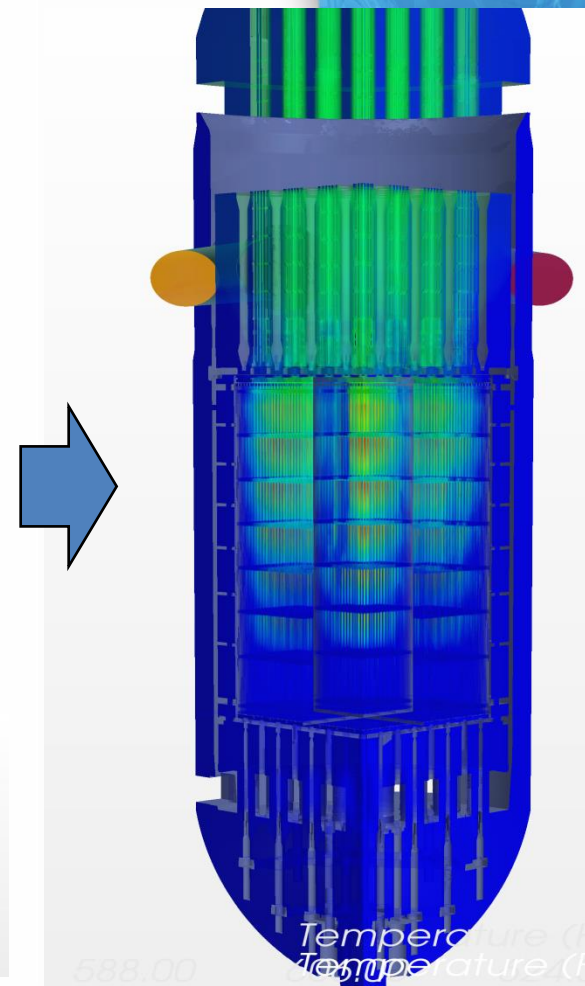
3D CAD



CFD



Mesh



Results

4-Loop Westinghouse PWR Multi-Physics Model Development

Benefits

- Local T&H conditions such as pressure, velocity, cross flow magnitude can be used to address challenge problems:
 - GTRF
 - FAD
 - GSI-191
- The design TH questions under normal operating and accident conditions such as:
 - Lower plenum flow anomaly
 - Pressure drop
 - Turbulence mixing coefficients input to channel code
 - Lift force
 - Cross flow between fuel assemblies
 - Bypass flow
- The local flow information can be used as boundary conditions for micro scale models.
- Local thermal conditions such as temperature and two phase flow can be used to address challenge problems:
 - CRUD/CILC
 - PCI
 - DNB
 - Cladding integrity during LOCA
 - AOA (Axial Offset Anomaly, boron hide out in crud)
- The normal design question can also be addressed:
 - Fuel center line temperature
 - Cladding hot spot
 - Gray rod center temperature
 - Thermal mixing at hot leg
- Provide Thermal boundary conditions for Micro scale models

